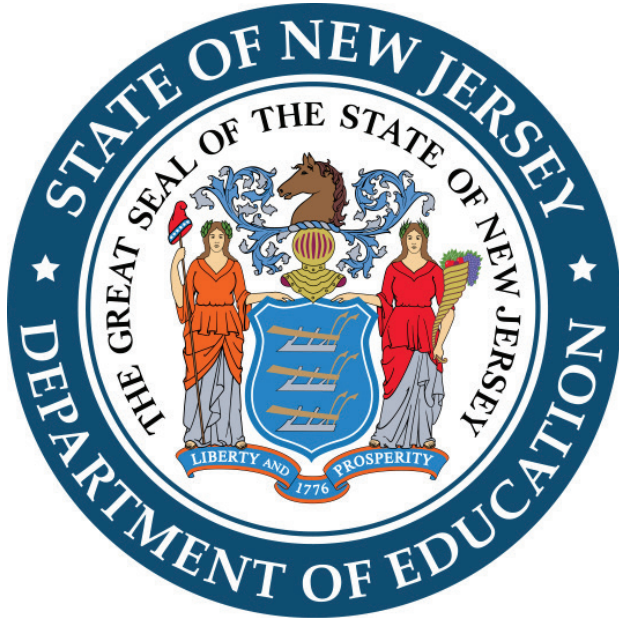




New Jersey
Student Learning
Assessments-Adaptive

Practice Test

Geometry

Student Name _____



Directions:

Today, you will take the Geometry Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
 - Write only one digit or symbol in each answer box.
 - Use only the options listed below each box.
 - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.

4	3							} Answer boxes
-	-	-	-	-	-	-	-	} Negative sign
.	.	.	.	.	.	.	.	} Decimal point
0	0	0	0	0	0	0	0	} Number bubbles
1	1	1	1	1	1	1	1	
2	2	2	2	2	2	2	2	
3	●	3	3	3	3	3	3	
●	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	



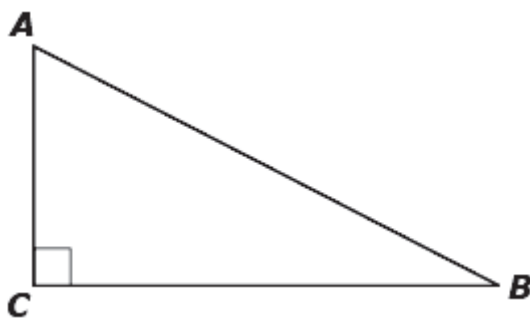
1. Hannah has a cone made of steel and a cone made of granite.
 - Each cone has a height of 10 centimeters and a radius of 4 centimeters.
 - The density of steel is approximately 7.75 grams per cubic centimeter.
 - The density of granite is approximately 2.75 grams per cubic centimeter.

What is the difference, to the nearest gram, of the masses of the cones?

−	−	−	−	−	−	−	−
•	•	•	•	•	•	•	•
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



2. Triangle ABC is shown.

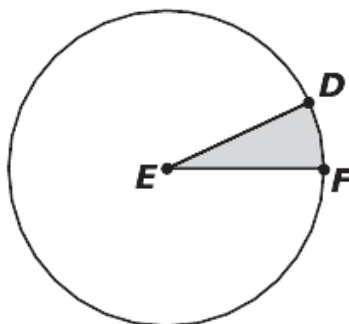


Which statement must be true?

- (A) $\cos(A) = \sin(A)$
- (B) $\cos(A) = \sin(B)$
- (C) $\cos(A) = \cos(B)$
- (D) $\sin(A) = \sin(B)$



3. Circle E is shown, where $\angle E$ measures 0.44 radians and $EF = 7$ inches.



What is the area of the shaded region, to the nearest hundredth of a square inch?

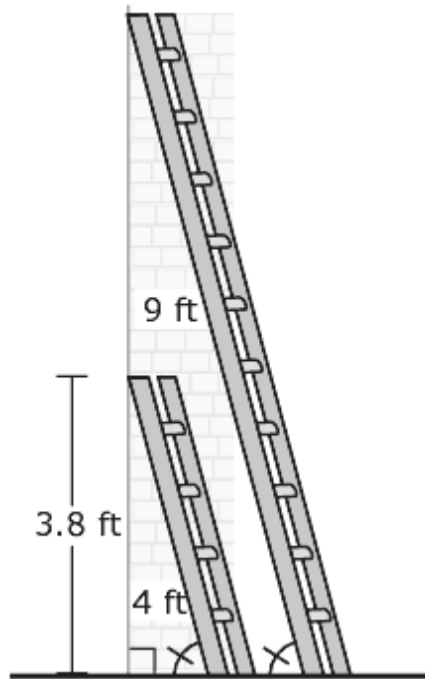
4. A square is rotated about its center.

Select all of the angles of rotation that will map the square onto itself.

- (A) 45 degrees
- (B) 60 degrees
- (C) 90 degrees
- (D) 120 degrees
- (E) 180 degrees
- (F) 270 degrees



5. A 9-foot (ft) ladder and a 4-foot ladder are leaning against a house. The two ladders create angles of the same measure with the ground. The 4-foot ladder has a height of 3.8 feet against the house.

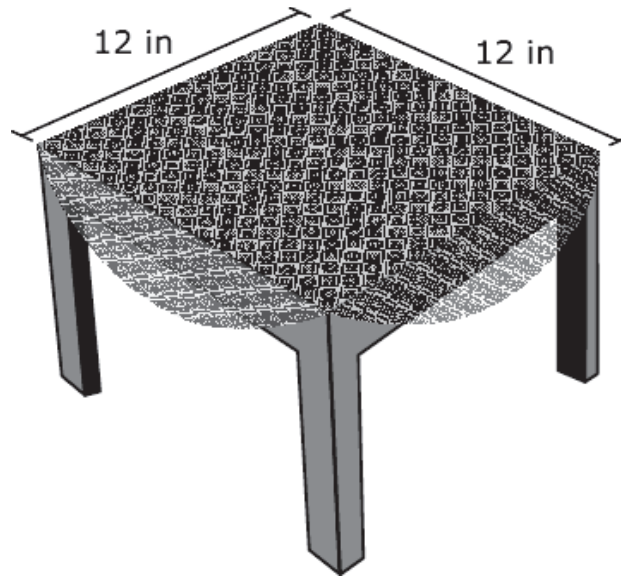


What is the height, in feet, of the 9-foot ladder against the house?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



6. Jackson has a table with a square top and he wants to buy a circular piece of lace that will cover the entire top of the table. The top of the table has side lengths of 12 inches, as shown.

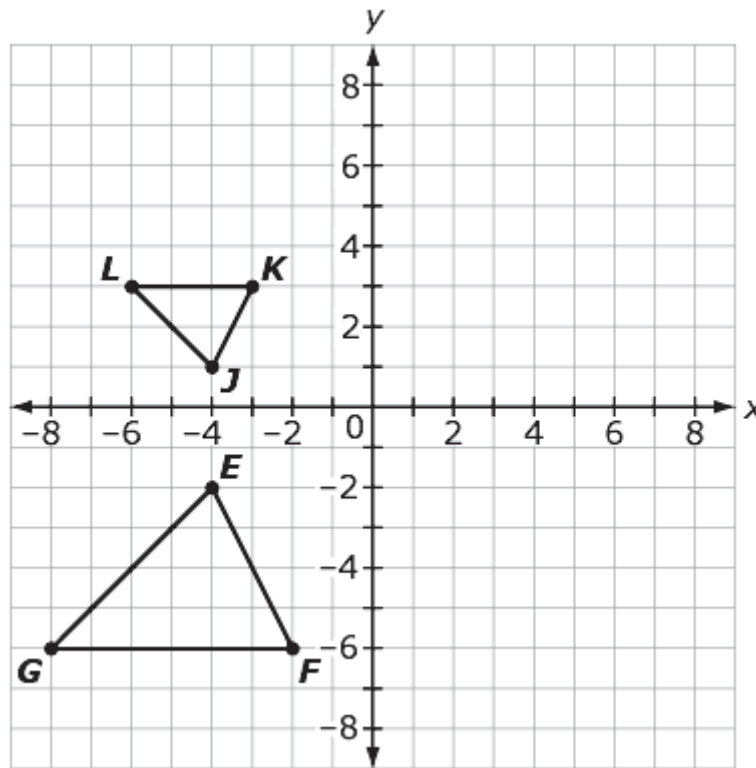


What is the area, in square inches, of the smallest circular piece of lace Jackson could buy? Round your answer to the nearest tenth.

−	−	−	−	−	−	−	−
•	•	•	•	•	•	•	•
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



7. Two triangles are shown.



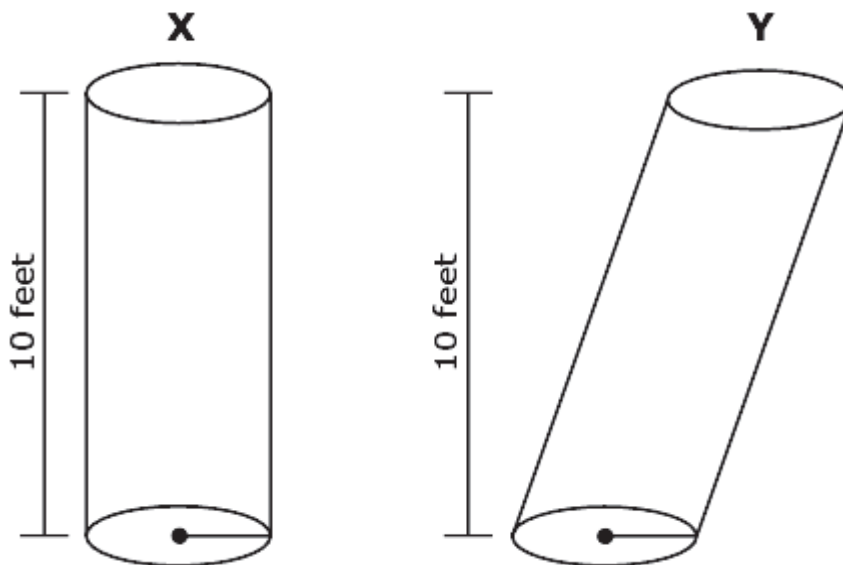
Which sequence of transformations could be performed on $\triangle EFG$ to show that it is similar to $\triangle JKL$?

- (A) Rotate $\triangle EFG$ 90° clockwise about the origin, and then dilate it by a scale factor of $\frac{1}{2}$ with a center of dilation at point F' .
- (B) Rotate $\triangle EFG$ 180° clockwise about point E , and then dilate it by a scale factor of 2 with a center of dilation at point E' .
- (C) Translate $\triangle EFG$ 1 unit up, then reflect it across the x -axis, and then dilate it by a scale factor of $\frac{1}{2}$ with a center of dilation at point E'' .
- (D) Reflect $\triangle EFG$ across the x -axis, then reflect it across the line $y = x$, and then dilate it by a scale factor of 2 with a center of dilation at point F'' .



8. What is the exact perimeter of a parallelogram with vertices at $(3, 2)$, $(4, 4)$ and $(6, 1)$?

9. Two cylinders, X and Y, are shown. Each cylinder has a height of 10 feet.

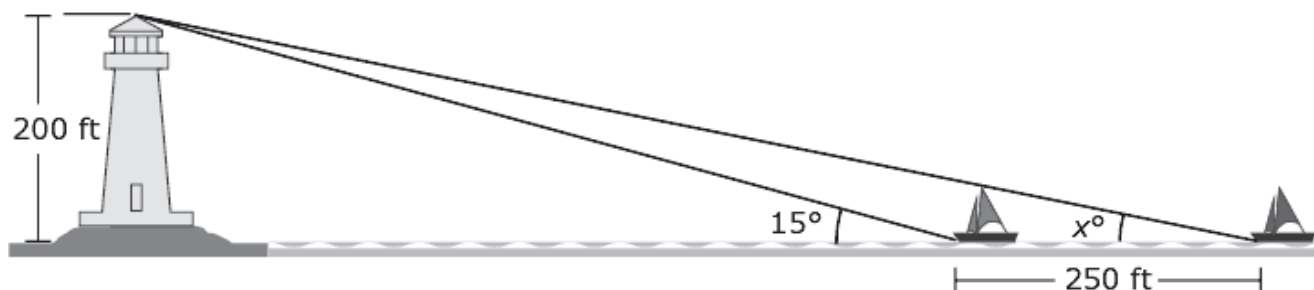


Which statement about these cylinders is true?

- (A) The volumes of the two cylinders are always equal because they have the same height.
- (B) The volume of cylinder Y is always greater because the slant height of cylinder Y is greater than the height of cylinder X.
- (C) The relationship between the volumes of the two cylinders cannot be determined because the slant height of cylinder Y is not given.
- (D) The relationship between the volumes of the two cylinders cannot be determined because the radii of the two cylinders are not given.



10. Two boats are traveling toward a lighthouse that is 200 feet (ft) above sea level at its top. When the two boats and the lighthouse are collinear, the boats are exactly 250 feet apart and the boat closest to the lighthouse has an angle of elevation to the top of the lighthouse of 15° , as shown.

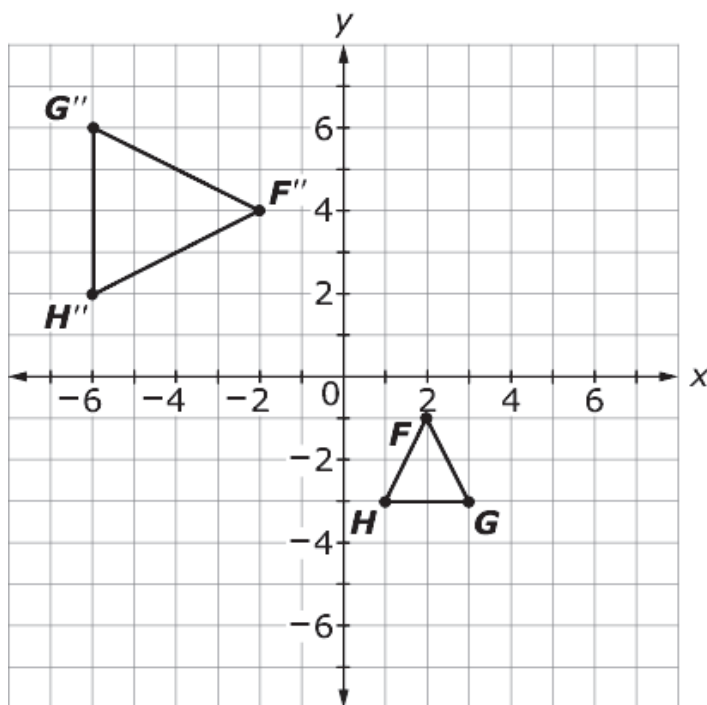


What is the value of x , rounded to the nearest hundredth?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



11. The coordinate plane shows $\triangle FGH$ and $\triangle F''G''H''$.



Which sequence of transformations can be used to show that $\triangle FGH \sim \triangle F''G''H''$?

- (A) a dilation about the origin with a scale factor of 2, followed by a 180° clockwise rotation about the origin
- (B) a dilation about the origin with a scale factor of 2, followed by a reflection over the line $y = x$
- (C) a translation 5 units up and 4 units left, followed by a dilation with a scale factor of $\frac{1}{2}$ about point F''
- (D) a 180° clockwise rotation about the origin, followed by a dilation with a scale factor of $\frac{1}{2}$ about point F''



- 12.** Steven constructs an equilateral triangle inscribed in circle P . His first three steps are shown.

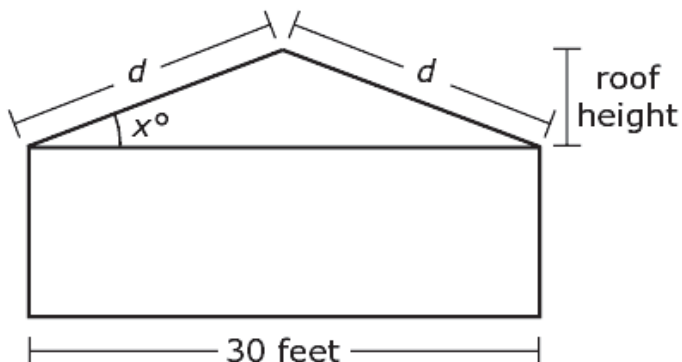
1. He creates radius $\overline{PQ}$ using a point Q on the circle.
2. Using point Q as the center and the length of $\overline{PQ}$ as a radius, he uses a compass to construct an arc that intersects the circle at R .
3. Using point R as the center and the length of $\overline{PQ}$ as a radius, he uses a compass to construct an arc that intersects the circle at S .

What should be Steven's next step in constructing the equilateral triangle?

- (A) Draw line segments connecting the points Q , R , and S to construct $\triangle QRS$.
 - (B) Draw line segments connecting the points P , R , and S to construct $\triangle PRS$.
 - (C) Construct an arc intersecting the circle by using point S as the center and the length of $\overline{PQ}$ as a radius.
 - (D) Construct an arc intersecting the circle by using point P as the center and the length of $\overline{PQ}$ as a radius.
- 13.** Create the equation of a line that is perpendicular to $2y = 14 + \frac{2}{3}x$ and passes through the point $(-2, 8)$.



14. Jeremy is building a garage, as shown. He wants the roof height to be between 3.5 and 5 feet. He must decide the angle measure to use for the pitch, or slant, of the roof when the slant height is d feet.

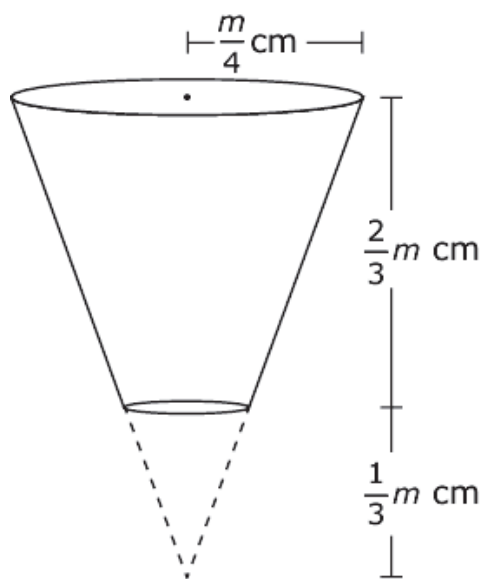


Which inequality can Jeremy use to ensure that his roof will be within the necessary height range?

- (A) $\frac{3.5}{30} \leq \tan(x) \leq \frac{5}{30}$
- (B) $\frac{3.5}{15} \leq \tan(x) \leq \frac{5}{15}$
- (C) $\frac{3.5}{15} \leq \sin(x) \leq \frac{5}{15}$
- (D) $\frac{3.5}{30} \leq \sin(x) \leq \frac{5}{30}$



15. A popcorn container is made from a cone with a portion of the cone removed and sealed, as shown. The removed portion of the cone has a height of $\frac{1}{3}m$.



Create an expression, using m , that represents the volume, in cubic centimeters, of the popcorn container.



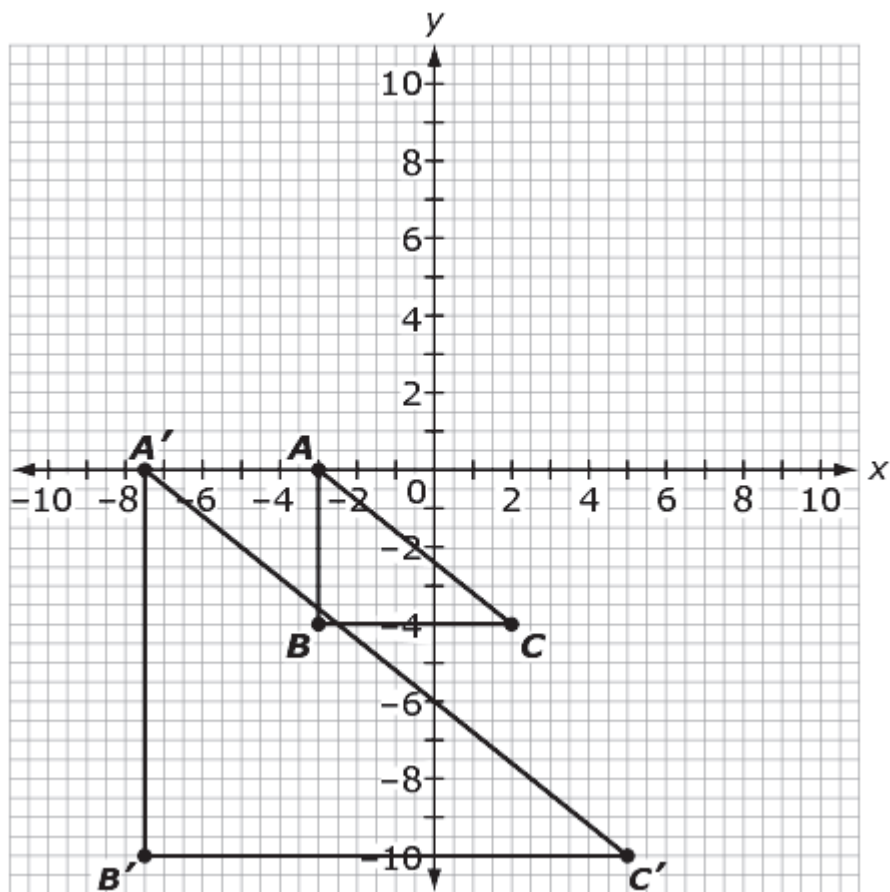
- 16.** All corresponding sides and angles of $\triangle RST$ and $\triangle DEF$ are congruent.

Select all of the statements that must be true.

- (A) There is a reflection that maps $\overline{RS}$ to $\overline{DE}$.
- (B) There is a dilation that maps $\triangle RST$ to $\triangle DEF$.
- (C) There is a translation followed by a rotation that maps $\overline{RT}$ to $\overline{DF}$.
- (D) There is a sequence of transformations that maps $\triangle RST$ to $\triangle DEF$.
- (E) There is not necessarily a sequence of rigid motions that maps $\triangle RST$ to $\triangle DEF$.



17. Triangle ABC is dilated with a scale factor of k and a center of dilation at the origin to obtain triangle $A'B'C'$.



What is the scale factor?

$k =$

-	-	-	-	-	-	-	-
.	.	.	.	.	.	.	.
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



18. Eric explains that all circles are similar. His argument with a missing step is shown.

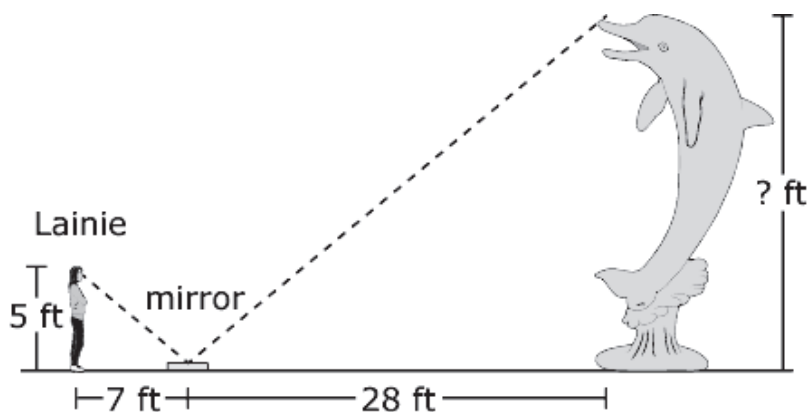
1. Let there be two circles, circle A and circle B.
2. There exists a translation that can be performed on circle A such that it will have the same center as circle B.
- 3.
4. Thus, there exists a sequence of transformations that can be performed on circle A in order to obtain circle B.
5. Therefore, circle A is similar to circle B.
6. Since circle A and circle B can be any circles, all circles are similar.

Which statement could be step 3 of the argument?

- (A) There exists a reflection that can be performed on circle A such that it will have the same radius as circle B.
- (B) There exists a dilation that can be performed on circle A such that it will have the same radius as circle B.
- (C) There exists a reflection that can be performed on circle B such that it will have the same center as circle A.
- (D) There exists a dilation that can be performed on circle B such that it will have the same center as circle A.



19. Lainie wants to calculate the height of a sculpture. She places a mirror on the ground so that when she looks into the mirror she sees the top of the sculpture, as shown.



What is the height, in feet, of the sculpture?

-	-	-	-	-	-	-	-
.	.	.	.	.	.	.	.
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9



Paper Practice Test Answer and Alignment Document: Math

Geometry

Item Sequence	Answer Key	Standards Alignment
1	any value between 837 and 838.1, inclusive	G.MG.A.2
2	B	G.SRT.C.7
3	10.78, or any equivalent value	G.C.B.5
4	C, E, F	G.CO.A.3
5	8.55, or any equivalent value	G.SRT.B.5
6	any value between 226 and 226.3, inclusive	G.MG.A.3
7	C	G.SRT.A.2
8	$2\sqrt{10} + 2\sqrt{5}$, or $2\sqrt{13} + 2\sqrt{5}$, or $2\sqrt{10} + 2\sqrt{13}$, or any equivalent expression	G.GPE.B.7
9	D	G.GMD.A.1
10	any value between 11.349 and 11.355, inclusive	G.SRT.C.8
11	B	G.SRT.A.2
12	C	G.CO.D.13
13	$y = -3x + 2$, or any equivalent equation	G.GPE.B.5
14	B	G.SRT.C.6
15	$\frac{1}{3}m\left(\frac{m}{4}\right)^2\pi - \frac{1}{3}\left(\frac{m}{12}\right)^2\left(\frac{m}{3}\right)\pi$, or any equivalent expression	G.MG.A.1
16	C, D	G.CO.B.7
17	2.5, or any equivalent value	G.SRT.A.1b
18	B	G.C.A.1
19	20, or any equivalent value	G.SRT.B.5